

Module 2 — Quiz A (Def & Notation)

Question bank · After Read 1 · open notes · unlimited attempts

Module 2, Quiz A: Definitions & notation

Reading: [Module 2 Read 1](#)

Project: [Project 2](#) · **Canvas:** Def & Notation quiz (Module 2)

Work these **practice** items after [Read 1](#). Sections follow Read 1: estimation vocabulary, population vs sample, binomial center/spread, and normal approximation. Section D is **optional** enrichment (Type I/II errors — not required for Project 2).

CI computation and R → [Synthesis](#) after the [exercise lab](#).

Section A — Estimation vocabulary (Read 1 §3–4)

A1. Population

Define **population** in one or two clear sentences.

A2. Sample

Define **sample** in one or two clear sentences.

A3. Parameter

Define **parameter** in one or two clear sentences.

A4. Statistic

Define **statistic** in one or two clear sentences.

A5. Sampling frame

Define **sampling frame** in one or two clear sentences.

A6. Bias

Define **bias** in the context of sampling and estimation.

A7. 95% confidence level

Define **95% confidence level** in terms of repeated sampling and intervals.

A8. Confidence interval

Define **confidence interval** in one or two clear sentences.

Answer key — Section A — Estimation vocabulary (Read 1 §3–4)

1. The **population** is the full group we want to learn about (the target group in the research question).
 2. The **sample** is the subset of population units we actually observe and collect data from.
 3. A **parameter** is a numerical summary of the population (usually unknown), such as μ or p .
 4. A **statistic** is a numerical summary computed from the sample, such as $\bar{x}$ or $\hat{p}$.
 5. The **sampling frame** is the list (or selection mechanism) used to choose the sample; it should closely match the population.
 6. **Bias** is a systematic tendency for an estimate to be too high or too low because of how data were sampled or measured.
 7. If we repeatedly sampled the same way and built 95% CIs each time, about **95%** of those intervals would contain the true parameter.
 8. A **confidence interval** is a range of plausible values for a population parameter, built from sample data using statistic $\pm$ margin of error.
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Section B — Population, sample, parameter, statistic

B1. Part-time job survey

Random sample $n = 120$ undergraduates; $x = 46$ work ≥ 10 hrs/week.

Identify population, sample, parameter, and statistic (compute $\hat{p}$).

B2. Classroom chairs

District $N = 300$ classrooms; SRS $n = 30$; $\bar{x} = 29.8$ chairs.

Identify population, sample, parameter, and statistic.

B3. Clinic satisfaction

$n = 80$ patients sampled; 52 very satisfied.

Identify population, sample, parameter, and statistic.

B4. Course website minutes

25 STAT 109 students; $\bar{x} = 18.4$ minutes/day on course site.

Identify population, sample, parameter, and statistic.

Answer key — Section B — Population, sample, parameter, statistic

1. **Population:** undergraduates of interest. **Sample:** 120 surveyed. **Parameter:** true proportion working ≥ 10 hrs/week (p). **Statistic:** $\hat{p} = 46/120 \approx 0.383$.
 2. **Population:** all 300 classrooms. **Sample:** 30 classrooms. **Parameter:** population mean chairs (μ). **Statistic:** $\bar{x} = 29.8$.
 3. **Population:** clinic patients of interest. **Sample:** 80 patients. **Parameter:** true proportion very satisfied (p). **Statistic:** $\hat{p} = 52/80 = 0.65$.
 4. **Population:** STAT 109 students of interest. **Sample:** 25 students. **Parameter:** population mean minutes (μ). **Statistic:** $\bar{x} = 18.4$.
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Section C — Binomial center & normal approximation (Read 1 §1–2)

C1. Binomial mean

ESP test: $n = 25$ trials, $p = 0.2$. What are μ and σ for $X = \text{number correct}$?

C2. Normal approximation conditions

A binomial count has $n = 100$, $p = 0.04$. Can you use the normal approximation with $np \geq 10$ and $n(1 - p) \geq 10$? Explain briefly.

C3. Empirical Rule

State the **Empirical Rule** for a normal distribution (three parts: 68%, 95%, 99%).

C4. Rejection region sketch

Under a normal null curve for a one-proportion test, where is the **rejection region** for a two-sided test at roughly $\alpha = 0.05$ using the Empirical Rule? How does shaded tail area relate to p-value?

Answer key — Section C — Binomial center & normal approximation (Read 1 §1–2)

1. $\mu = np = 5$. $\sigma = \sqrt{np(1-p)} = \sqrt{25(0.2)(0.8)} = 2$.
 2. $np = 4$ and $n(1-p) = 96$. **No** — $np < 10$, so normal approximation is not appropriate.
 3. About **68%** within $\pm 1\sigma$, **95%** within $\pm 2\sigma$, **99%** within $\pm 3\sigma$ of the mean.
 4. Reject in **both tails** beyond about $\pm 2\sigma$ from the center. Total shaded tail area approximates the **p-value** (probability of a result at least as extreme under H_0).
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Section D — Type I/II vocabulary (optional enrichment)

Optional — not required for Project 2 (CI focus). Helpful if you study hypothesis testing later.

D1. Significance level

Define the **significance level** α in hypothesis testing.

D2. Type I error

Define a **Type I error** in hypothesis testing.

D3. Type II error

Define a **Type II error** in hypothesis testing.

Answer key — Section D — Type I/II vocabulary (optional enrichment)

1. α is the maximum acceptable probability of a **Type I error** and the cutoff for rejecting H_0 (reject when p-value $< \alpha$).
 2. Rejecting H_0 when H_0 is actually true (false alarm).
 3. Failing to reject H_0 when H_0 is actually false (missed detection).
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